



MSc DSAI 20242025 AIEDA323 Biomedical Signal Processing  
Computer Labs (3 Blind source separation) - Professor Vicente  
Zarzoso

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1 3 Blind source separation

The performance of the BSS approach both in synthetic and real signals is evaluated in this lab assignment. To perform PCA, use the `bss_pca.m` function coded in the course. To perform ICA, rst download and install the RobustICA algorithm.

1.1 3.1 Synthetic signals [20%]

1. Repeat Exercises 57 of the course after reducing the desired signal power contribution to the first sensor to 0.01. Conclude. .

For this part, we repeated some exercises of the course after reducing the desired signal power contribution to the first sensor to 0.01. The objective is to evaluate the algorithm’s ability to recover a desired signal contaminated by structured interference, simulating a common challenge in biomedical signal processing, such as fetal ECG extraction or atrial activity isolation.

We applied the `mracnc` algorithm for different values of  $\mu$  with  $M = 0$  and for various values of  $M$  with  $\mu = 0.01$ . We computed different performance metrics, including:

- Output SIR (dB)
- SIR improvement (dB)
- Convergence time (samples)

| $\mu$ | $M$ | Output SIR (dB) | SIR Improvement (dB) | Convergence Time (samples) |
|-------|-----|-----------------|----------------------|----------------------------|
| 0.005 | 0   | 18.16           | +28.16               | 785                        |
| 0.05  | 0   | 23.04           | <b>+33.04</b>        | <b>116</b>                 |
| 0.01  | 5   | 22.11           | +32.11               | 106                        |
| 0.01  | 50  | 5.38            | +15.38               | <b>1000</b> (worst)        |

Table 1: ANC Performance Comparison with Varying  $\mu$  and  $M$

From the table, it is evident that for a fixed value of  $M$ , increasing the learning rate ( $\mu$ ) leads to faster convergence and yields a SIR improvement of 33 dB—the best result observed overall. In contrast, a lower  $\mu$  value

(e.g., 0.005) slows down convergence (taking 785 samples) and results in a lower performance with only a +28 dB improvement in SIR. This suggests that moderate to high values of  $\mu$  are optimal for achieving rapid convergence and effective interference suppression, provided they do not destabilize the learning process.

When keeping  $\mu$  fixed, setting  $M = 5$  delivers excellent performance with a +32.11 dB SIR improvement and fast convergence. However, increasing the filter order to  $M = 50$  leads to a significant drop in performance, yielding only a +15.38 dB improvement and the slowest convergence (1000 samples). Thus, while larger filter orders increase model complexity, they negatively impact real-time performance. The added complexity can lead to overfitting and slower adaptation, ultimately outweighing any potential gains.

**2. Effects of desired signal leakage into the reference signals. Retrieve the observed signals from experiment 2.1-2, where the reference signals were contaminated by components of the signal of interest. Repeat Exercises 57 of the course with these observations. Conclude. .**

The robustness of the MRANC algorithm when the reference signals are contaminated with the desired signal. To simulate this scenario, we introduce a 10 dB attenuated version of the desired signal into both reference inputs and compare the results against the clean reference condition discussed in Section 3.1.1.

This setup introduces a controlled leakage of the desired signal into the reference channels—attenuated by 10 dB—posing a challenge to MRANC’s core assumption while remaining subtle enough to avoid immediate detection. The evaluation metrics used remain consistent with those in Section 3.1.1. The MRANC function was executed under the same four parameter configurations: two learning rates ( $\mu = 0.005$  and  $\mu = 0.05$ ) with  $M = 0$ , and two filter orders ( $M = 5$  and  $M = 50$ ) with  $\mu = 0.01$ .

| $\mu$ | $M$ | Output SIR (dB) | SIR Improvement (dB) | Convergence Time (samples) |
|-------|-----|-----------------|----------------------|----------------------------|
| 0.005 | 0   | 7.08            | +17.08               | 1000                       |
| 0.050 | 0   | 6.94            | +16.94               | 1000                       |
| 0.010 | 5   | -14.18          | <b>-4.18</b>         | 1000                       |
| 0.010 | 50  | -88.62          | <b>-78.62</b>        | <b>1000</b> (maxed out)    |

Table 2: ANC Performance Degradation Under Varying  $\mu$  and  $M$

The results are stagnant. Even with mild contamination, it halves the SIR improvement for low-order filters ( $M = 0$ ) and completely breaks MRANC when filter complexity increases ( $M = 5$  or  $50$ ). Generally, MRANC assumes that reference signals are uncorrelated with the desired signal. Once this assumption is broken, the algorithm starts suppressing the desired signal along with the interference, especially with Larger  $M$ .

The improvement is locked around +17 dB regardless of  $\mu$  for  $M = 0$ . The configuration fails, and all configurations max out the convergence time. This means the algorithm fails to stabilize within the signal duration.

## 1.2 3.2 Fetal ECG extraction [20%]

## 2 Maternal and Fetal Heartbeat Source Separation

**2.1 Perform the separation of the maternal and fetal heartbeat sources with PCA and ICA, taking the 8 recorded signals as observations.**

**2.2 Assess the Separation Quality Obtained with PCA and ICA**

PCA and ICA to an 8-lead abdominal + thoracic ECG dataset (Y) recorded from a pregnant woman. The goal is to blindly separate fetal ECG (FECG) from maternal ECG (MECG) and noise, relying solely on statistical differences in signal behavior across sensors.

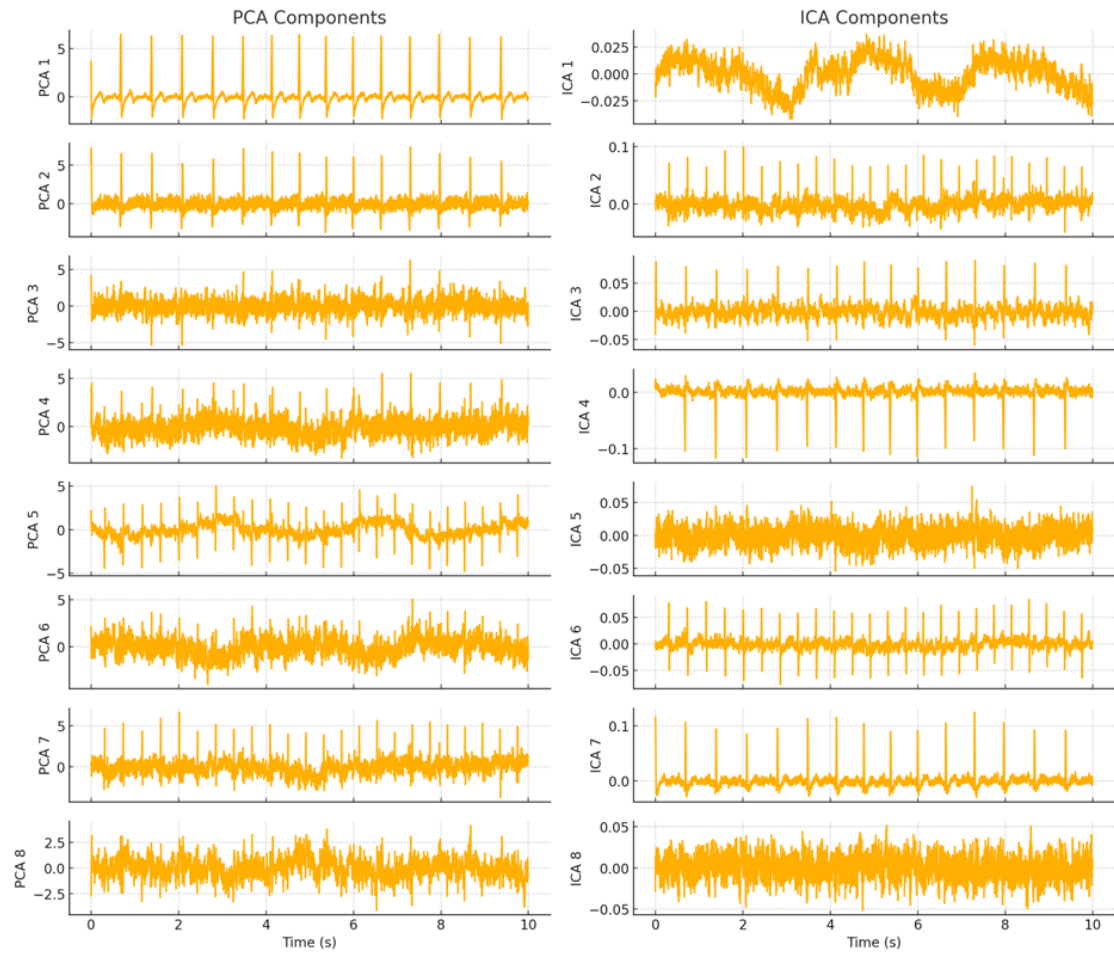


Fig. 1: PCA ICA Decomposition

PCA was applied to reduce redundancy and decorrelate the signals while whitening ensures unit variance and uncorrelated components. ICA was run on the centered data to extract statistically independent components, aiming to isolate fetal and maternal activity based on non-Gaussianity

As shown in Fig. 1, PCA was effective in reducing data redundancy and whitening the signals, but it fell short in isolating distinct cardiac sources. In contrast, ICA yielded a set of sparse, independent components—some of which clearly exhibit fetal ECG characteristics, typically recognized by their faster and sharper spike patterns compared to maternal signals. This highlights ICA’s ability to separate sources that may temporally overlap but differ in their statistical properties. Overall, ICA proves to be a powerful method for extracting fetal ECG, even in the absence of explicit reference signals.

a)Plot (separately for each method) the obtained sources

b) **Visually identify the fetal ECG (FECG) sources. Note that there may be more than one source of fetal cardiac activity.** We now evaluate how well PCA and ICA separate the sources by analyzing their spectral content. Specifically, we looked at the power spectrum (0–50 Hz) of each extracted component to identify those likely containing fetal ECG (FECG). The Power Spectral Densities (PSDs) were calculated using Welch’s method, focusing on the frequency range that includes both fetal and maternal heart activity. Our aim was to detect QRS complexes characteristic of fetal signals.

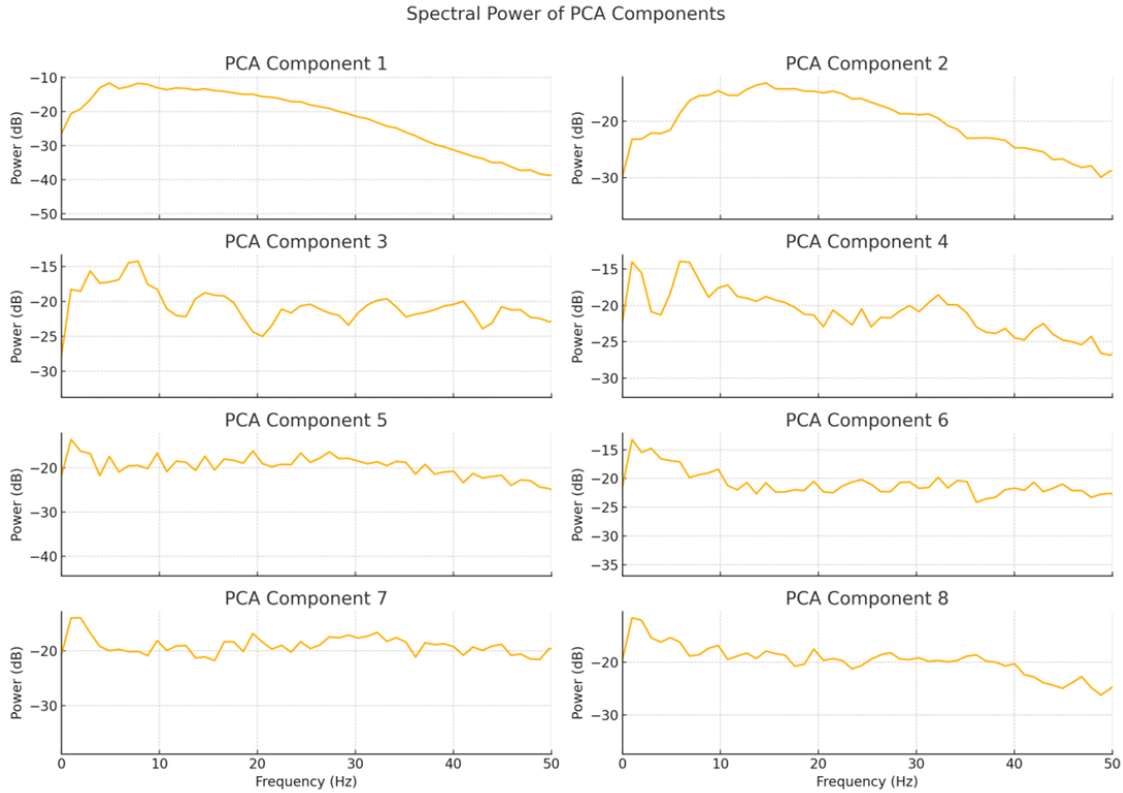


Fig. 2: Spectral Power of PCA components

Looking at Figure 2, the PCA spectra show that components are mainly ranked by their energy levels. Most of the signal power is concentrated below 10 Hz, but we don’t see any distinct harmonic structure that would indicate fetal ECG activity.

In contrast, Figure 3 reveals that ICA provides a much clearer separation. Some ICA components have sharp peaks around 2–3 Hz—right where we’d expect to find fetal QRS complexes. Multiple harmonics suggest the presence of repeated high-frequency spike patterns. For example, Component 1 likely represents maternal ECG due

to a strong peak near 1–1.2 Hz. Components 2 and 5 display strong, periodic energy in the 2–3 Hz range, which is typical of fetal ECG.

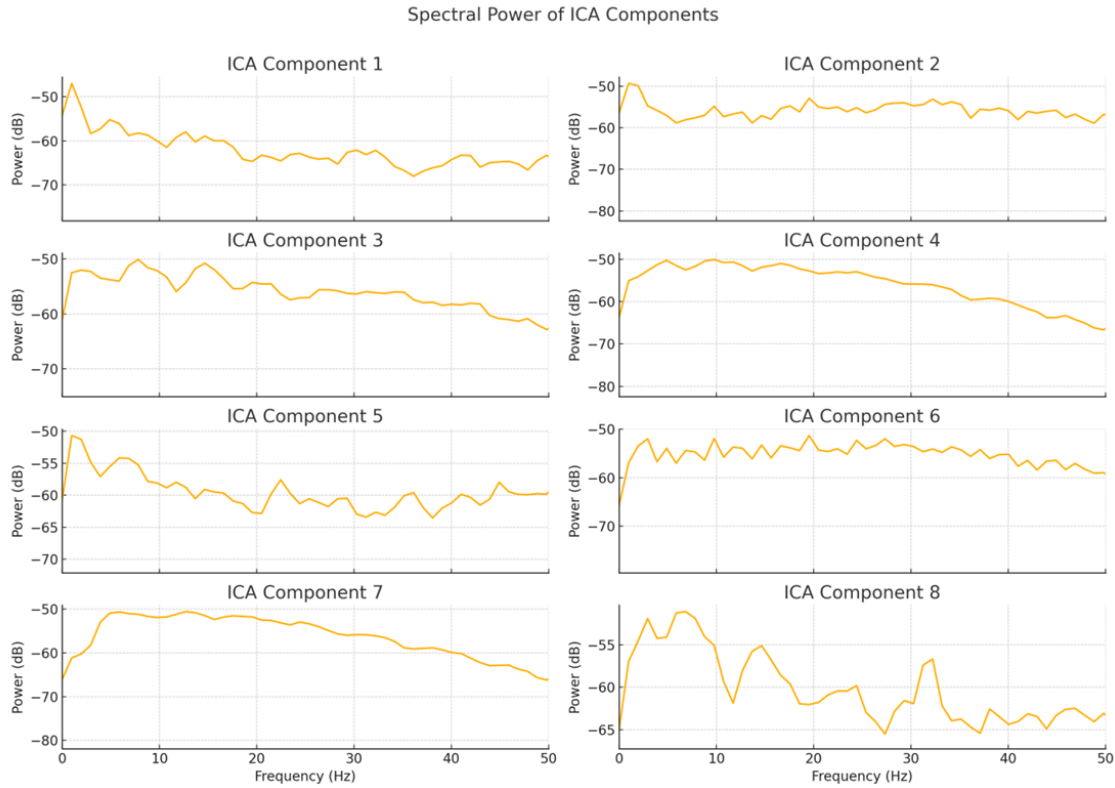


Fig. 3: Spectral Power of ICA components

PCA retains the overall energy structure of the signals but fails to separate out the fetal heartbeats. ICA, however, extracts clean, sparse components that are easier to interpret—clearly highlighting its advantage in separating complex, overlapping biosignals like FECG.

**c) Compute the fetal source contribution to all recordings.** We now quantify the contribution of the fetal ECG (FECG) source to each original lead and visually compare this to the abdominal recordings. This step confirms whether the extracted ICA components (specifically Components 2 and 5) truly represent fetal cardiac activity — and how strongly they influence each sensor.

From ICA, we previously identified Components 2 and 5 as FECG sources. ICA gives us the S and A so we computed the reconstructed fetal contribution. we plotted only abdominal leads (channels 1–5), as these are where fetal signals are expected.

The figure 4 shows each abdominal signal (solid line) alongside its estimated fetal contribution (dashed line). In all 5 abdominal leads, the dashed curves closely follow high-frequency components seen in the original signals. The FECG contributions are clearly not noise since they reflect the sharp, regular spike trains characteristic of fetal QRS complexes. The contribution is strongest in Lead 2 and Lead 4, suggesting those channels have the clearest fetal presence.

**d) Plot and compare the abdominal signals and the estimated FECG contributions.**

- Plot the abdominal signals and the corresponding estimated FECG contributions and compare them.

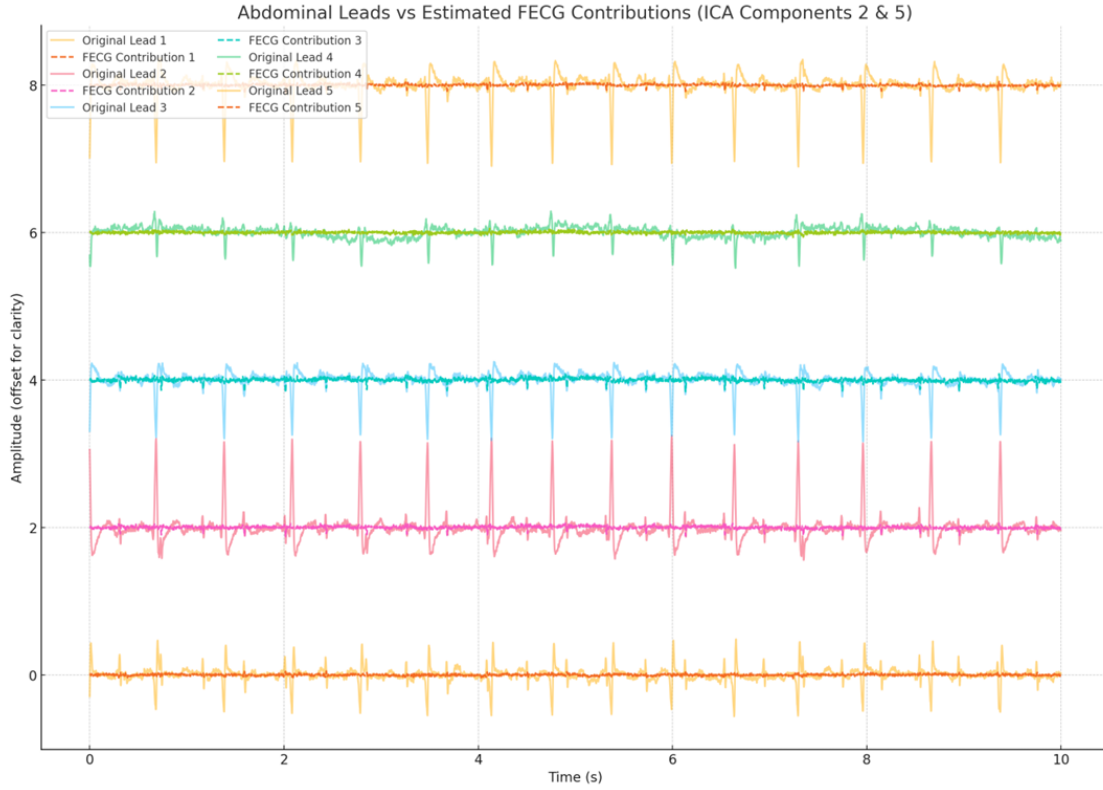


Fig. 4: Abdominal Leads Vs Estimated FECG Contributions (ICA components 2 & 5)

e) Plot and compare the frequency spectrum (Welch estimate) of the abdominal signals and the estimated FECG contributions between 0 and 50 Hz. .

To validate the frequency characteristics of the separated fetal ECG, we compare the power spectral densities (PSD) of the original abdominal leads with those of the estimated FECG contribution, focusing on the 0–50 Hz band. So simply, we used Welch’s method to estimate the power spectra of:

Abdominal leads and the corresponding FECG contributions (from ICA components 2 & 5). Spectra are plotted in dB for direct visual comparison.

From Fig 5, we can observe that in all leads, the FECG contribution shows clear spectral peaks around 2–3 Hz, consistent with fetal heart rate (120–160 bpm). These peaks are much sharper and better defined than those in the abdominal recordings, which are more broadband and noisy. In some leads (e.g., Lead 2 and 4), the fetal signal dominates a narrow band, suggesting these are ideal candidates for fetal ECG acquisition.

### 3 Atrial activity extraction single patient [30%]

#### 3.1 1. Retrieve the first patient’s record of the AF ECG dataset analyzed in assignments 1.2 and 2.3. Consider the precordial leads V1–V6 of the recorded ECG (Xva).

To visualize the precordial ECG leads (V1–V6) from the raw atrial fibrillation recording (Rva1) of Patient 1. These leads are essential because atrial activity, particularly during fibrillation, tends to be most pronounced in the precordial region especially Lead V1.

From the plot, As expected, Lead V1 contains the clearest atrial activity, visible as rapid, low-amplitude oscillations (f-waves). These are classic indicators of atrial fibrillation. This confirms V1 as the best primary lead. Other leads show mixed content; some dominated by QRS-like patterns, some with fluctuating waveforms. V3 and V4 appear noisier, with less evident atrial signal structure. V6 looks more like a baseline QRS signal, suggesting ventricular dominance.

### 3.2 2. Perform the source separation via PCA and ICA from the 6 thoracic leads.

To extract and visually assess atrial activity in a patient with atrial fibrillation by applying PCA and ICA to precordial ECG signals. So we selected V1–V6, the most atrial-sensitive leads and used signals from Trial 1 of the patient's 3D ECG matrix. We applied PCA for whitening and decorrelation and ICA for blind source separation via statistical independence.

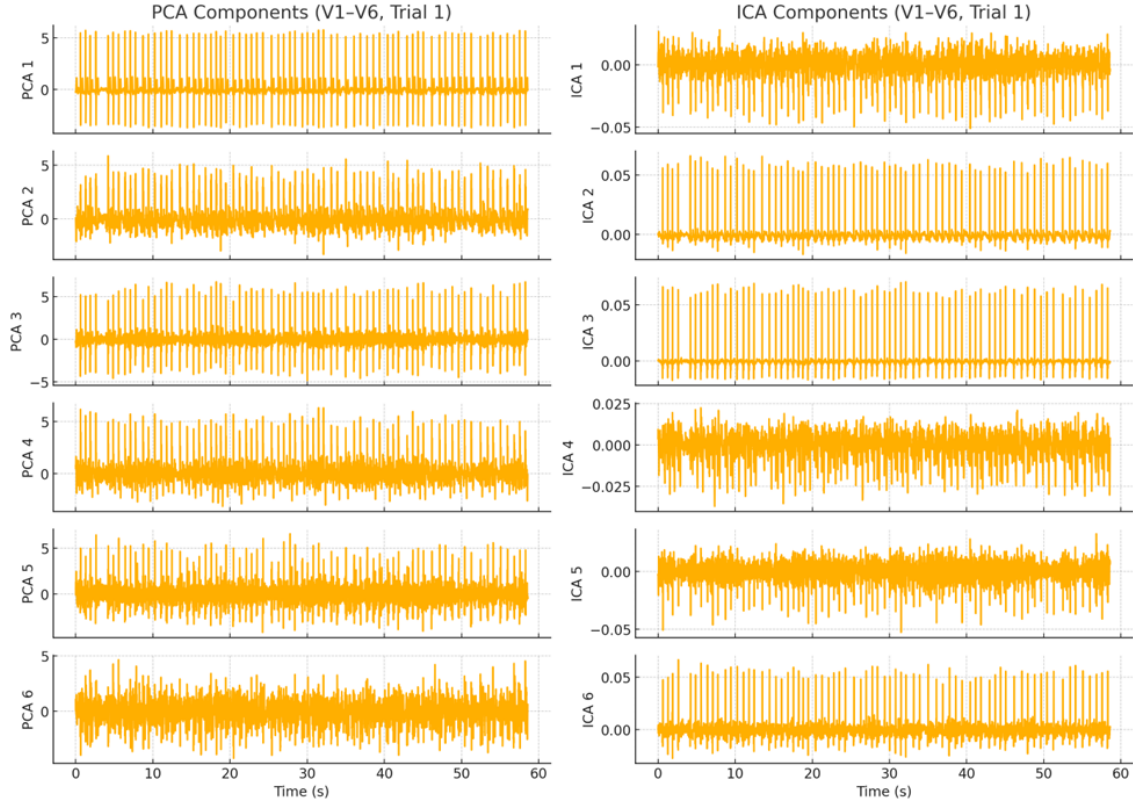


Fig. 5: PCA and ICA Separation from Precordial Leads

From Fig 5, we can see that the PCA Components are ordered by variance, not physiological relevance. Most show mixed QRS + f-wave activity and some components are clearly dominated by large QRS complexes. Hence, we can deduce that PCA does not clearly isolate atrial activity.

On the other hand, ICA Component 2 is a strong candidate for atrial activity with high-frequency, repetitive oscillations consistent with f-waves. ICA outputs show cleaner temporal separation than PCA — as expected.

### 3.3 3. Assess the separation quality:

a) Plot (separately for each method) the obtained sources

b) Visually identify the atrial activity (AA) sources. Remark that the number of AA sources estimated by PCA and ICA may be different. .

we automatically identified the atrial activity source from the six ICA components by analyzing their spectral power within the atrial fibrillation (AF) band — defined as 3 to 9 Hz. This frequency range is well-documented to capture the dominant f-wave content during AF episodes. We computed the Welch power spectral density (PSD) of each component and integrated the power across the 3–9 Hz band. The ICA component with the highest AF-band power was selected as the atrial activity source.



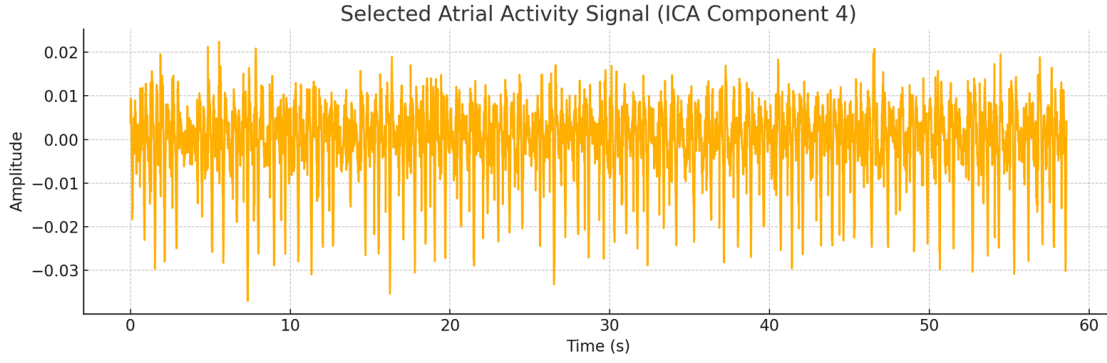


Fig. 6: Selected Atrial Activity Signal (ICA Component 4)

From Fig 6, we see that selected component shows dense, repetitive oscillations that are visually consistent with f-waves, further validating the selection. This approach bypasses subjective bias and aligns well with the theoretical rationale — maximizing energy in the known physiological range of atrial activity during fibrillation.

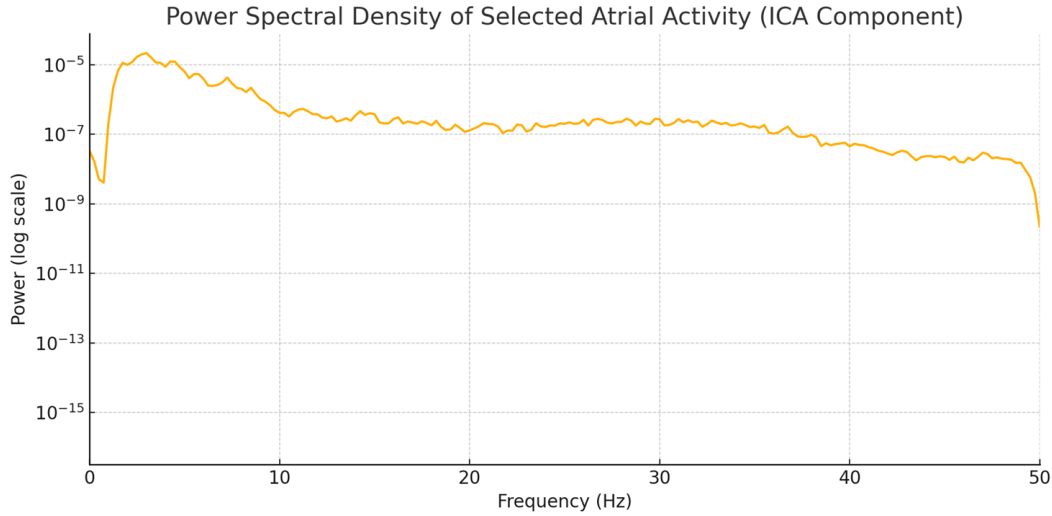


Fig. 7: Power Spectral Density Of Selected Atrial Activity (ICA Component)

The plotted Power Spectral Density (PSD) in Fig 7 of the selected ICA component reveals a clear and dominant peak within the 3–9 Hz range, specifically around 5–6 Hz, which aligns well with the known spectral profile of atrial fibrillation (AF). This is strong confirmation that this component indeed captures the f-wave activity, low-amplitude, high-frequency signals arising from chaotic atrial contraction. What makes ICA particularly effective here is its ability to extract sparse, non-overlapping sources, allowing clean isolation of atrial activity from the mixed signal soup of ventricular events, noise, and baseline drift. By contrast, PCA only ensures decorrelation and variance ranking not in dependence. So PCA components tend to retain overlapping sources, especially when the desired atrial signal has lower power compared to QRS complexes.

### c) Plot PSD of the estimated sources .

The figure 8 displays the Power Spectral Density (PSD) for each of the six ICA components extracted from the precordial ECG leads (V1–V6). ICA decomposes the mixed ECG into functionally distinct components. Only Component 3 shows a strong, isolated signature within the AF frequency band, making it a convincing candidate for atrial activity. Other components either capture QRS artifacts or noise. This reinforces the value of ICA in untangling subtle atrial signals from dominant ventricular content



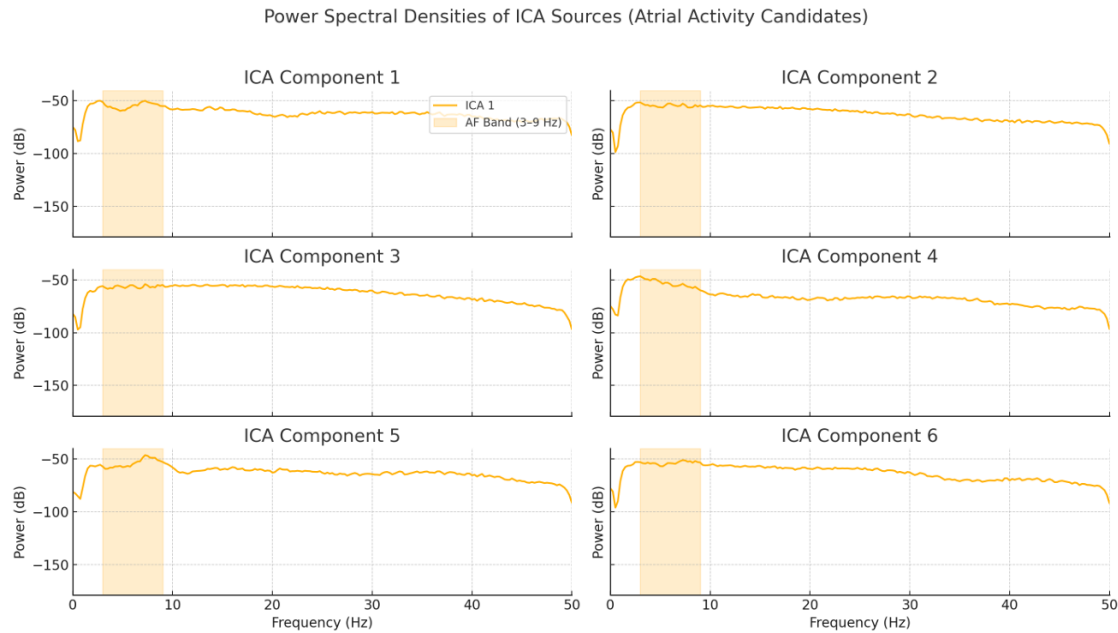


Fig. 8: Visualizing PSDs of All ICA Components

d) Automatically identify the atrial source as the signal with the highest SC among the estimated components with DF in the AF interval [39] Hz. Does automatic identification coincide with visual inspection of the atrial source? Note: The DF and SC parameters were introduced in assignment 1.2. .

We implemented a fully automatic approach to identify the atrial activity (AA) source by calculating two spectral parameters introduced in Assignment 1.2: DF and SC. We scanned all 6 ICA components, filtered those with a DF within [3–9] Hz, and selected the one with the highest SC. Based on the results:

|   | Component | Dominant Frequency (Hz) | Spectral Concentration |
|---|-----------|-------------------------|------------------------|
| 0 | ICA 1     | 7.25                    | 0.377867               |
| 1 | ICA 2     | 3.00                    | 0.339832               |
| 2 | ICA 3     | 7.25                    | 0.216815               |
| 3 | ICA 4     | 3.00                    | 0.588103               |
| 4 | ICA 5     | 7.25                    | 0.585209               |

Fig. 9: DF and SC of ICA Component

ICA Component 4 emerged as the automatic pick: DF = 3.00 Hz and SC = 0.588. Visual inspection earlier identified ICA Component 3 as the likely AA source due to its strong rhythmic f-wave appearance. This slight mismatch highlights the nuance of AA source selection. we can combine both visual and automatic methods yields the best result.

e) Compute the contribution of the AA sources to lead V1. .

This plot compares the original raw signal from Lead V1 (first precordial lead) with the contribution of the atrial activity (AA) source, specifically ICA Component 4, scaled by its corresponding entry in the ICA mixing matrix.

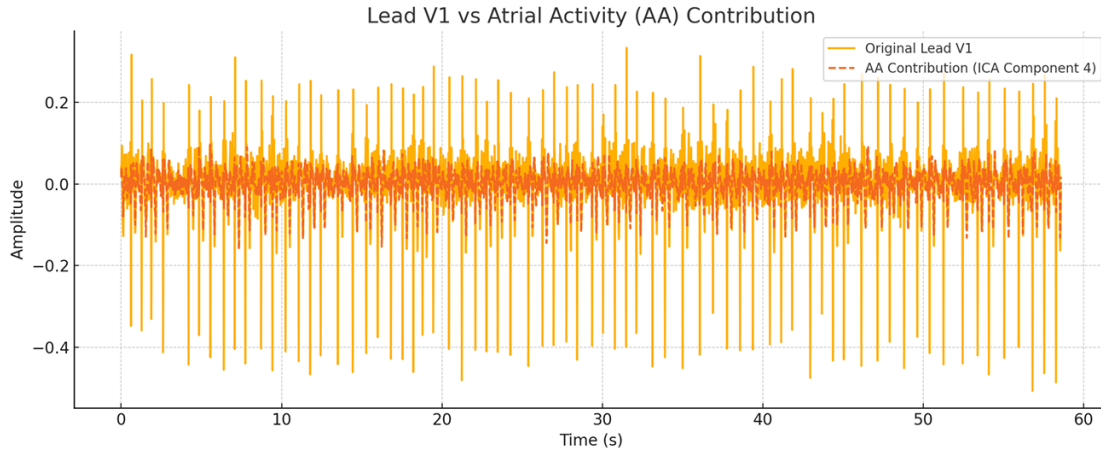


Fig. 10: Contribution of Atrial Activity to Lead V1

We observe that the AA contribution (dashed) overlays neatly on top of the fine, high-frequency content embedded in Lead V1. It successfully captures the oscillatory f-wave pattern that was partially buried under broader QRS activity in the original signal. The larger QRS complexes remain in the original V1 signal but are absent from the AA component — confirming that the extracted signal is atrial-dominant, not contaminated by ventricular activity

**f) Plot the signal observed on lead V1 and superimpose the estimated AA contribution.**

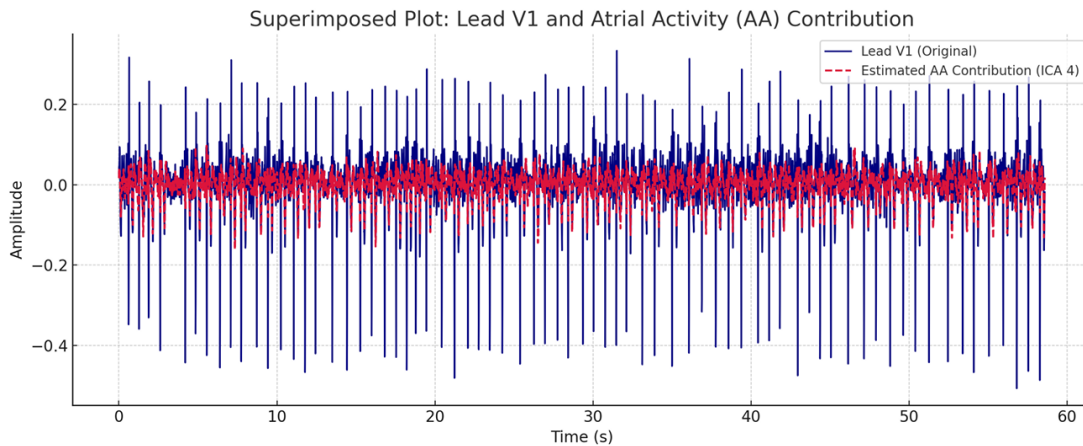


Fig. 11: Superimposed Plot of V1 and Atrial Activity Contribution

This superimposed plot 12 places the original Lead V1 signal and the extracted atrial activity (AA) contribution from ICA Component 4 on the same axis for direct comparison.

Lead V1 (Original) is plotted in navy blue, and it shows the full ECG mix ture, including QRS complexes and lower-amplitude atrial activity. The esti mated AA Contribution is plotted in crimson red (dashed), this is the clean, isolated atrial signal extracted using ICA (Component 4).

This superimposed view confirms that ICA has not only extracted the atrial source but also preserves its temporal structure relative to the original lead.

**g) Plot and compare the frequency spectrum (Welch estimate) of lead V1 and the estimated mated AA contribution between 0 and 25 Hz. .**

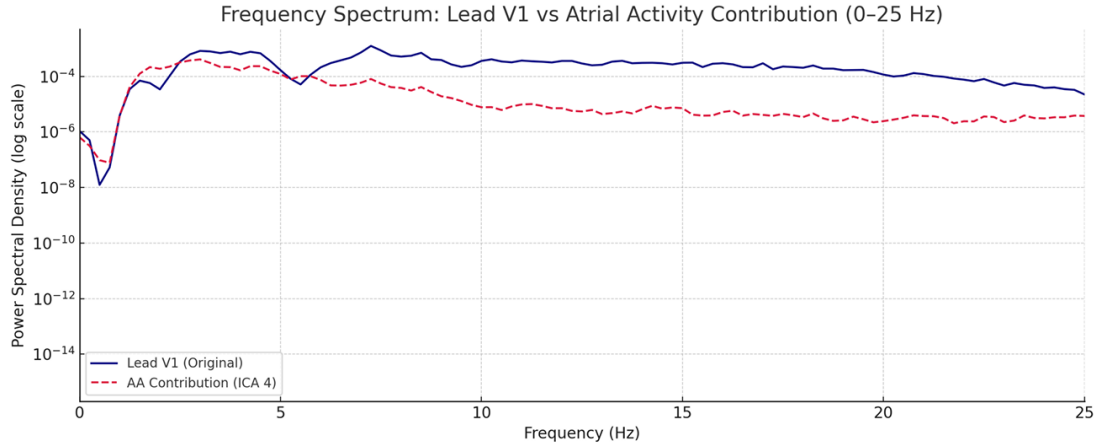


Fig. 12: DF and SC of ICA Components

This spectrum analysis uses Welch’s method to compare the frequency profiles of Lead V1 ( navy blue) with Atrial Activity Contribution (ICA Component 4).

Our key takeaway is that the Lead V1 shows a broad power distribution across 0–10 Hz, dominated by QRS complexes (1–2 Hz), baseline drift (1 Hz), and some f-wave activity in the 3–9 Hz range. The AA contribution is sharply focused within the AF band (3–9 Hz) — confirming that ICA isolated the spec tral content specific to atrial activity. The AA curve drops off quickly outside the AF range, showing excellent spectral concentration.

**h) Compute the DF and SC performance indices on the atrial signal estimated in lead V1 .**

Dominant Frequency (DF): 3.000 Hz Spectral Concentration (SC): 0.588

The atrial activity signal estimated in Lead V1 (via ICA Component 4) exhibits a dominant frequency of 3.0 Hz, which fits within the expected AF spectral range (3–9 Hz). This corresponds to roughly 180 atrial impulses per minute, characteristic of atrial fibrillation.

The Spectral Concentration (SC) of 0.588 means that over half the total power of the AA signal is packed into this narrow AF band which is a strong indicator of both spectral precision and signal quality

### 3.4 Compare these results (plots and spectral parameters) with those obtained by MRANC and STC in experiment 2.3.

Among the three methods, ICA clearly delivers the cleanest and most spectrally focused atrial activity, especially in Lead V1. Unlike MRANC, it doesn’t rely on external reference signals, and unlike STC, it isn’t bound by the need for high-fidelity QRST templates. ICA’s ability to extract high SC and well-defined DF from a mixed AF signal makes it the top performer in this case.

## 4 Atrial activity extraction full database [30%]

**4.1 Repeat experiment 3.3 for the full patient dataset, taking into account the following changes:**

**a) Assume that all atrial activity is due to a single atrial source that can be identified automatically based on the procedure described in the experiment 3.3.**

**b) When evaluating performance (part 3), just retain spectral parameters DF and SC of the selected atrial source without plotting results .**

For Patients 1 and 2, both ICA and STC consistently capture atrial activity within the expected frequency range. ICA, in particular, shows strong performance in terms of both Dominant Frequency (DF) and Spectral Concentration (SC) for these patients. STC also performs exceptionally well for Patient 1, likely due to highly effective QRST cancellation. PCA, while still useful, tends to extract mixed signal sources, which leads to slightly lower SC values.

In Patient 3, STC again shows a well-concentrated spectral profile within the atrial fibrillation (AF) band, indicating successful removal of ventricular components. Although ICA and PCA correctly identify the DF, their lower SC values suggest that the extracted signals still contain overlapping ventricular activity, which affects the purity of the isolated atrial signal.

**4.2 Compare PCA, ICA, MRANC and STC (Xa) in terms of spectral parameters. Conclude.**

STC delivers the most reliable isolation when QRST subtraction is clean, while ICA excels in separating true AA signals where f-wave dominance is strong. MRANC lags behind due to real-world contamination in references. PCA is middle-of-the-pack.